WAYGROUND formerly Quizizz **Worksheets**

Name

Class

Date

Cloud Architecting - Week 7

Total questions: 21

Worksheet time: 21mins

1. Why is AWS more economical than traditional data centers for applications with varying compute workloads?

   a) Amazon EC2 costs are billed on a monthly basis.
   b) Customers retain full administrative access to their Amazon EC2 instances.
   c) Customers can permanently run enough instances to handle peak workloads.
   d) Amazon EC2 instances can be launched on-demand when needed.

2. If your project requires monthly reports that iterate through very large amounts of data, which Amazon Elastic Cloud (Amazon EC2) purchasing option should you consider?

   a) Spot Instances
   b) Scheduled Reserved Instances
   c) Dedicated Hosts
   d) On-Demand Instances

3. What is included in an Amazon Machine Image (AMI)?

   a) A template for the root volume for the instance
   b) Launch permissions that control which AWS accounts can use the AMI to launch instances.
   c) A block device mapping that specifies volumes to attach to the instance when it's launched
   d) All of the above

4. Which Amazon Elastic Compute Cloud (Amazon EC2) feature ensures your instances will not share a physical host with instances from any other customer?

   a) Amazon VPC
   b) Placement groups
   c) Dedicated Instances
   d) Reserved instances

5. Which of the following services is a serverless compute service in AWS?

    a) AWS Config
    b) AWS Lambda
    c) AWS OpsWorks
    d) Amazon EC2

6. What is the service provided by AWS that enables developers to easily deploy and manage applications in the cloud?

    a) Amazon Elastic Container Service
    b) AWS Elastic Beanstalk
    c) AWS Opswork
    d) AWS CloudFormation

7. Your web application needs four instances to support study traffic all of the time. On the last day of the month, the traffic triples. What is the most cost-effective way to handle this pattern?

    a) Run 12 Reserved Instances all of the time.
    b) Run four On-Demand Instancees constantly, then add eight more On-Demand instances on the last day of each month.
    c) Run four Reserved Instances constantly, then add eight On-Demand Instances on the last day of each month.
    d) Run four On-Demand Instances constantly, then add eight Reserved Instances on the last day of each month.

8. Containers contain an entire operating system.

    a) True
    b) False

9. Which Amazon EC2 option is best for long-term workloads with predictable usage patterns?

    a) Spot Instances
    b) On-Demand Instances
    c) Reserved Instances

10. Which of the following must be specified when launching a new Amazon Elastic Compute Cloud (Amazon EC2) Windows instance?

    a) Password for the administrator account
    b) The Amazon EC2 instance ID
    c) Amazon EC2 instance type
    d) Amazon Machine Image (AMI)

11. Low Choon Keat is a good lecturer

    a) Yes
    b) No

12. Which definition describes a virtual private cloud (VPC)?

a) A virtual private network (VPN) in AWS Cloud

b) An extension of an on-premises network into AWS

c) A logically isolated virtual network that you define in the AWS Cloud

d) A fully managed service that extends the AWS Cloud to customer premises

13. Which actions are best practices for designing a virtual private cloud (VPC)? (Select THREE.)

a) Divide the VPC network range evenly across all Availability Zone available

b) Use the same CIDR block as your on-premises network.

c) Create one subnet per Availability Zone for each group hosts that have unique routing requirements.

d) Reserve some address space for future use.

14. A company wants to run a highly available web tier by using two EC2 instances and a load balancer. Which design is valid and provides the highest availability?

a) One subnet in one Availability Zone. The subnet contains two EC2 instances.

b) One subnet, which spans two Availability Zone. Each Availability Zone contains one instance.

c) Two different subnets in the same Availability Zone. Each subnet contains one EC2 instance.

d) Two different subnets, one per Availability Zone. Each subnet contains one EC2 instance.

15. A company's VPC has the CIDR block 172.16.0.0/21 (2048 address). It has two subnets (A and B). Each subnet must support 100 usable address now, but this number is expected to rise to at most 254 usable addresses soon. Which subnet addressing scheme meets the requirements and follow AWS best practices?

a) Subnet A: 172.16.0.0/25 (128 addresses)
Subnet B: 172.16.0.128/25 (128 addresses)

b) Subnet A: 172.16.0.0/25 (128 addresses)
Subnet B: 172.16.0.128/25 (128 addresses)

c) Subnet A: 172.16.0.0/23 (512 addresses)
Subnet B: 172.16.2.0/23 (512 addresses)

d) Subnet A: 172.16.0.0/22 (1024 addresses)
Subnet B: 172.16.4.0/22 (1024 addresses)

16. Which combination of actions enables direct internet access for IPv4 hosts in a virtual private cloud (VPC)? (Select THREE.)

a) Configuring security groups and network ACLs to permit internet traffic

b) Configure a default route that points to the virtual private gateway

c) Configuring hosts to have or obtain an internet-routable address

d) Creating a route for 0.0.0.0/0 that points to the internet gateway

17. A group of consultants requires access to an EC2 instance from the internet, for 3 consecutive days each week. The instance is shut down the rest of the week. The virtual private cloud (VPC) has internet access. How should you assign an IPv4 address to the instance to give the consultants access?

a) Associate an elastic IP address with the EC2 instance

b) Enable automatic address assignment for the subnet

c) Enable automatic address assignment for the EC2 instance

d) Assign the IP address in the operating system (OS) boot configuration

18. Several EC2 instances launch in a VPC that has internet access. These instances should not be accessible from the internet, but they must be able to download updates from the internet. How should the instances launch?

a) With Elastic IP address, in a subnet with a default route to an internet gateway

b) With public IP addresses, in a subnet with a default route to an internet gateway

c) Without public IP addresses, in a subnet with a default route to an internet gateway

d) Without public IP addresses, in a subnet with a default route to a NAT gateway

19. You are configuring a bastion host to access EC2 instances in a virtual private cloud (VPC). What must you do to the security groups? (Select TWO.)

a) Add a rule to the bastion host to deny all the traffic from the internet.

b) Add a rule to the bastion host to allow traffic from your source IP address.

c) Add a rule to the bastion host to allow return traffic to your source IP address.

d) Add a rule to the private subnet EC2 instances to allow traffic from the bastion host security group.

20. You have a virtual private cloud (VPC) with a public subnet and a secure subnet. All EC2 instances in the secure subnet must be able to communicate with specific internet addresses. How can you control traffic with a network ACL?

a) Add rules to the default network ACL to allow traffic from and to allowed internet addresses.

b) Add rules to the default network ACL to allow traffic from and to allowed internet addresses. Deny all other traffic.

c) Add rules to the subnet custom network ACL to allow traffic from and to allowed internet addresses.

d) Add rules to the subnet custom network ACL to allow traffic from and to allowed internet addresses. Deny all other traffic.

21. All of the EC2 instances in a subnet can communicate with a certain IPv4 network on the internet. How should you modify the security groups or current custom network ACL to deny traffic to and from several restricted address in that network?

a) In the network ACL, deny traffic to and from the restricted addresses.

b) In the security group, deny traffic to and from the restricted addresses.

c) In the network ACL, allow traffic only to and from address ranges that excluded the restricted addresses.

d) In the security group, allow traffic only to and from address ranges that excluded the restricted addresses.

Answer Keys

1. d) Amazon EC2 instances can be launched on-demand when needed.
2. b) Scheduled Reserved Instances
3. d) All of the above
4. c) Dedicated Instances
5. b) AWS Lambda
6. b) AWS Elastic Beanstalk
7. c) Run four Reserved Instances constantly, then add eight On-Demand Instances on the last day of each month.
8. b) False
9. c) Reserved Instances
10. d) Amazon Machine Image (AMI), c) Amazon EC2 instance type
11. a) Yes
12. c) A logically isolated virtual network that you define in the AWS Cloud
13. a) Divide the VPC network range evenly across all Availability Zone available, c) Create one subnet per Availability Zone for each group hosts that have unique routing requirements.
14. a) Reserve some address space for future use, d) Two different subnets, one per Availability Zone. Each subnet contains one EC2 instance.
15. c) Subnet A: 172.16.0.0/23 (512 addresses) Subnet B: 172.16.2.0/23 (512 addresses)
16. a) Configuring security groups and network ACLs to permit internet traffic, c) Configuring hosts to have or obtain an internet-routable address
17. a) Creating a route for 0.0.0.0/0 that points to the internet gateway, d) Associate an elastic IP address with the EC2 instance
18. d) Without public IP addresses, in a subnet with a default route to a NAT gateway

19. b) Add a rule to the bastion host to allow traffic from your source IP address., d) Add a rule to the private subnet EC2 instances to allow traffic from the bastion host security group.

20. c) Add rules to the subnet custom network ACL to allow traffic from and to allowed internet addresses.

21. a) In the network ACL, deny traffic to and from the restricted addresses.